CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2017 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Repeater operates in which layer of the OSI model? Explain its function.

Explanation: A **Repeater** operates at the **Physical Layer (Layer 1)** of the OSI model. It receives weak or distorted electrical/optical signals from one cable segment, cleans, amplifies, regenerates the signal to its original strength and retimes it before transmitting it over the next segment, overcoming transmission attenuation.

Q2 (3M) Differentiate between FDM and TDM Multiplexing.

Differences: FDM divides available channel bandwidth into distinct non-overlapping frequency bands (analog). TDM allocates the entire channel bandwidth to each source for dedicated repeating time slots (digital).

Q3 (3M) A local telephone network is an example of which switching technology?

Answer: Circuit Switching. A dedicated physical communication circuit is established between the calling and called parties through telephone exchanges for the entire duration of the call.

Q4 (3M) What is Guided Media? List examples.

Definition: Physical solid transmission media that provide a physical conduit for electromagnetic signals: Twisted Pair Cable, Coaxial Cable, and Fiber Optic Cable.

Q5 (3M) Compare TCP and UDP at the Transport Layer.

Comparison: TCP is connection-oriented, reliable (acknowledgments, retransmissions), with flow and congestion control (HTTP, FTP). UDP is connectionless, unreliable, lightweight, and low-latency (DNS, VoIP, video streaming).

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain Seven Layers of ISO/OSI Reference Model.

OSI Architecture: Physical, Data Link, Network, Transport, Session, Presentation, Application layers with primary duties.

Q7 (7.5M) Explain Stop-and-Wait Flow Control Protocol and calculate its efficiency.

Efficiency: \eta = \frac{T_t}{T_t + 2T_p} = \frac{1}{1 + 2a} where a = \frac{\text{Propagation Time}}{\text{Transmission Time}}.

Q8 (7.5M) Explain Dijkstra's Algorithm for Shortest Path Routing.

Dijkstra's Algorithm: Iteratively selects minimum-cost unvisited nodes to build the shortest-path routing tree.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) Explain Error Control using CRC with complete mathematical division example.

CRC Division: Modulo-2 polynomial division with verification at receiver side.

Q10 (15M) **Explain Sliding Window Protocols: Go-Back-N and Selective Repeat.**

Sliding Window: In-depth operation, sliding window mechanics, and timeout handling.

Q11 (15M) **Explain Transmission Media: UTP, STP, Coaxial, and Optical Fiber.**

Detailed Media: Physical construction, connectors (RJ45, BNC, SC/ST), bandwidth, and propagation speed.

Q12 (15M) **Explain Congestion Control Algorithms: Leaky Bucket and Token Bucket.**

Traffic Shaping: Comparison of uniform leak rate vs burst absorption using tokens.

Q13 (15M) **Explain Switching Techniques: Circuit, Packet, and Message Switching with detailed diagrams.**

Switching Architectures: Latency, store-and-forward delay, and buffer utilization.
